

ComicCraft – Completed Project Phase Documentation

Compiled from the supplied ComicCraft 8-page project report and aligned to the provided AI/ML & Gen-AI project templates.

Project	ComicCraft – AI-Powered Comic Story Generator
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Department	Computer Science with Data Science
Institution	Shri Krishnaswamy College for Women
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1. Brainstorming & Ideation

Define Problem Statements

- Problem Statement: Users may have story ideas but lack the writing, illustration, character-design, panel-arrangement, and formatting skills or time required to turn those ideas into illustrated comics.
- Customer viewpoint: A user wants to provide a story prompt and customization details such as character name, setting, tone, and art style, then receive a structured visual comic.
- Pain addressed: Text-generation and image-generation tools are often separate; ComicCraft brings the stages into one workflow.

1. Brainstorming & Ideation

Empathy Map

- Says/asks: User provides a story prompt and preferences for character, setting, tone, and art style.
- Thinks: Wants a simple way to transform an idea into a visual story without handling every production stage manually.
- Does: Enters the prompt, submits the form, previews the generated comic, and exports it as a PDF.
- Feels/desired outcome: The report states the goal is to make comic creation simpler and more accessible for students, storytellers, and creative users.

1. Brainstorming & Ideation

Brainstorming & Idea Prioritization

- Core idea: AI-powered web application for personalized comic story generation.
- Key capabilities identified in the report: structured story generation, detailed narration/dialogue, AI illustrations, five-panel layout, preview, and PDF export.
- Prioritization basis used here: features explicitly stated as project objectives/features in the supplied report.

2. Requirement Analysis

Solution Requirements

- Functional requirements: collect user inputs; generate a structured comic outline; generate narration/dialogue; generate panel illustrations; assemble a comic layout; provide preview; export PDF.
- Integration requirements: FastAPI backend, Gemini models for text generation, Stable Diffusion for image generation, HTML/CSS/Jinja2 for UI and rendering.
- Non-functional expectations stated or implied by the report: simple interface, modular architecture, responsive web interface, and maintainability.

2. Requirement Analysis

Customer Journey Map

- 1) User has a comic idea → 2) enters prompt and customization → 3) backend validates request → 4) AI generates story structure and detailed text → 5) images are generated → 6) panels are assembled → 7) user previews → 8) PDF is exported.

2. Requirement Analysis

Data Flow Diagram

- Input: story prompt + character + setting + tone + art style.
- Flow: Browser/UI → FastAPI backend → Gemini Flash (outline) → Gemini Pro (narration/dialogue) → Stable Diffusion (panel images) → Layout Builder → Preview → PDF Export.
- Output: completed five-panel comic and downloadable PDF.

Architecture/Data flow summary: UI → FastAPI → Gemini text generation + Stable Diffusion image generation → Layout → Preview → PDF.

2. Requirement Analysis

Technology Stack

- Python — core application logic and AI workflow.
- FastAPI — backend framework, requests/routing/generation workflow.
- Google Gemini Flash — structured comic outline.
- Google Gemini Pro — detailed narration and dialogue.
- Stable Diffusion — comic-style panel illustrations.
- HTML/CSS — frontend presentation.
- Jinja2 — dynamic comic preview templates.
- PDF generation library/tool — downloadable comic documents.

3. Project Design Phase

Solution Architecture

- Layered architecture: User Interface Layer → FastAPI Backend Layer → Story Generation Layer and Image Generation Layer → Layout Layer → PDF Export Layer.
- The backend coordinates the AI generation process and the generated text/images are assembled into the final comic.

Architecture/Data flow summary: UI → FastAPI → Gemini text generation + Stable Diffusion image generation → Layout → Preview → PDF.

3. Project Design Phase

Proposed Solution

- Provide a single web application that converts user-defined story information into a structured, illustrated five-panel comic and downloadable PDF.
- The solution integrates text generation, image synthesis, layout management, preview, and document generation in one workflow.

3. Project Design Phase

Problem-Solution Fit

- Problem: comic production normally requires separate writing, illustration, character design, panel arrangement, and formatting stages.
- Solution fit: ComicCraft unifies these stages through one application and uses Generative AI to reduce manual effort while keeping user-provided story details as inputs.

4. Project Planning Phase

Project Planning

- Planning flow reflected in the report: define scope and objectives → select technology stack → design architecture/workflow → implement modules → integrate AI services → preview and PDF export → document results.
- The supplied report does not specify individual dates, task owners, or a detailed project schedule; those fields are therefore not invented here.

5. Project Development Phase

Coding & Solution

- Implementation uses a FastAPI backend, HTML/CSS/Jinja2 interface, Gemini-based story generation, Stable Diffusion image generation, layout assembly, preview, and PDF export.
- Module sequence: interface → FastAPI → story generation → image generation → layout builder → PDF export.

5. Project Development Phase

Code-Layout, Readability and Reusability

- Modular separation is explicitly described: each major task is developed and maintained independently.
- Reusable boundaries include UI collection, backend coordination, story generation, image generation, layout building, and PDF export.
- The report does not provide actual source-code files or a code-style checklist, so detailed code-level claims are not added.

5. Project Development Phase

No. of Functional Features Included in the Solution

- 1. AI-based comic story generation from prompts.
- 2. Character/setting/scene/art-style customization.
- 3. Automatic five-panel comic generation.
- 4. AI-generated illustrations per panel.
- 5. Dynamic comic preview.
- 6. PDF export.
- 7. FastAPI-based backend.
- 8. Modular architecture.

6. Project Testing

Performance Testing

- The report describes an end-to-end workflow and notes that AI service availability and response time can affect generation.
- Validated workflow described: request handling → story generation → illustration generation → layout → preview → PDF export.
- No numerical latency, throughput, load-test results, or benchmark values are provided in the supplied report, so none are fabricated here.

7. Project Documentation

Sample Project Documentation

- The supplied 8-page report itself provides the project documentation structure: Introduction; Problem Statement/Objectives/Features; Technology Stack; Architecture/Workflow/Data Flow; Module Description; Implementation/Results; Advantages/Limitations/Future Scope; Conclusion/References.

7. Project Documentation

Project Executable Files

- The report documents the application as a web-based FastAPI project and describes its runtime workflow.
- The supplied materials do not include a runnable source-code package or executable artifact; therefore this section records the documented application components rather than claiming an executable was supplied.

8. Project Demonstration

Project Demo Planning

- Demo sequence: open home page → enter story prompt and customization → submit → show generated story structure/content → show generated panel illustrations → show five-panel preview → export/download PDF.
- Demo should emphasize the complete user-to-output workflow described in the report.

8. Project Demonstration

Communication

- Presentation points: project motivation, problem addressed, objectives, technology stack, architecture, workflow, generated comic preview, results, limitations, and future scope.
- Team members named in the supplied report: Roshini P, Asrin Fathima M Y, Sivashree S.

8. Project Demonstration

Scalability & Future Plan

- Future scope explicitly listed: user login/authentication; cloud storage; multilingual comic generation; additional themes and visual styles; voice input; real-time editing; mobile application versions; improved AI models and generation quality.
- Current limitations: single comic generation, no authentication, no cloud storage, quality dependent on models/prompts, and AI-service availability/response time can affect generation.

8. Project Demonstration

Demonstration of Proposed Features

- Demonstrate prompt-based story generation, customization, five-panel generation, AI illustrations, dynamic preview, and PDF export.
- The supplied report states that the described implementation demonstrates an end-to-end Generative AI workflow transforming a textual idea into a multi-panel visual comic.

8. Project Demonstration

Team Involvement in Demonstration

- Team: Roshini P; Asrin Fathima M Y; Sivashree S.
- The supplied report does not assign specific demonstration roles to individual members. Individual role allocation is therefore left unspecified rather than invented.

Source Alignment Note

This compiled document is based on the supplied ComicCraft 8-page report. Where a template requested information that the report did not provide (for example, exact dates, numerical performance benchmarks, source-code listings, or individual team roles), the gap is explicitly noted rather than filled with unsupported details.

The original template PDFs are preserved alongside this completed compilation in the final ZIP package.